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Project Report:

BPAlign: A Machine Learning–Based Assistant for Hypertension Detection in Healthcare and Research

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This is to certify that the MSc project report titled:

"BPAlign: A Machine Learning–Based Assistant for Hypertension Detection in Healthcare and Research"

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BPAlign: A Machine Learning–Based Assistant for Hypertension Detection in Healthcare and Research

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Abstract

Hypertension is a major global health problem, especially in rural areas where regular checkups are rare. Telemedicine platforms collect large amounts of patient data, but these often lack verified outcomes, which limits their use for machine learning. We present BPAlign, a web-based tool to help clinicians and researchers assess hypertension risk. BPAlign is trained on the Framingham dataset and allows users to enter patient data to get risk predictions. It connects standard datasets with real-world telemedicine practice. Future versions will support feedback system and automatic data intregation according to feedback.

1 Introduction

BPAlign is not just another prediction tool—it is a window into the "why" behind predictions. It allows Telemedicine practitioners and researchers to compare models, spot anomalies, and quickly gain insights from new datasets, all within the critical context of hypertension screening. Its core mission is to transform raw patient data—from telemedicine records or hospital databases—into actionable intelligence.

Hypertension remains a leading cause of cardiovascular disease worldwide. For telemedicine practitioners, early identification of patients at high risk can be a game-changer. Platforms like the Dhaka University Telemedicine Program (DUTP), which serve rural communities where regular health checkups are rare, often encounter minor health issues that go unnoticed.

BPAlign serves both as a decision-support system for clinicians and as a research platform for exploring predictive models in rural healthcare contexts. For example, by analyzing DUTP's telemedicine records, BPAlign provides insights into rural populations' hypertension risk as a public health concern. It can not only predict individual hypertension risk but also offer valuable insights for healthcare providers and researchers who have health records of particular areas or patients with specific trends. This dual functionality—supporting both clinical decision-making and data-driven research—positions BPAlign as a practical and research-ready tool for advancing preventive care in underserved populations.

1.1 Motivation

The motivation behind BPAlign arises from practical challenges in translating machine learning models into clinical decision support systems. The key issues, problems, and BPAlign solutions are summarized below:

- **Issue: Out-of-sample performance**
  Problem: How well does the model perform on data beyond its training sample?
  Solution: Addressed by *Out-of-Sample Prediction*, where models trained on the Framingham dataset are tested on DUTP rural telemedicine data.

- **Issue: Model disagreement**
  Problem: What happens when different models provide conflicting predictions?
  Solution: Handled through the *Disagreement Cases* feature, which identifies and highlights mismatched outputs across models.

- **Issue: Feature influence**
  Problem: Which variables most strongly drive the predictions?
  Solution: Provided via the *Feature Importance* tab, allowing users to inspect key drivers for each model.

- **Issue: Generalization**
  Problem: Can the model trained on a benchmark dataset (e.g., Framingham) provide reliable predictions for rural telemedicine data (e.g., DUTP)?
  Solution: Demonstrated through the *Model Generalization* feature, which explicitly evaluates cross-dataset applicability.

- **Issue: Transparency**
  Problem: How can predictions and results be presented in a way that clinicians and researchers can interpret and trust?
  Solution: Supported by *ROC Curve Visualization*, detailed *Model Reports*, and human-readable diagnostic comments.

These issues and solutions bridge the gap between achieving high performance on controlled datasets and ensuring interpretability, trust, and applicability in real-world healthcare settings, particularly for rural telemedicine.

1.2 Literature Review

Hypertension is a leading risk factor for cardiovascular disease (CVD) worldwide. Early detection and prediction are critical for preventing complications such as stroke, heart attack, and kidney failure. With the expansion of telemedicine, integrating predictive algorithms into remote healthcare platforms has become increasingly relevant. This section reviews key studies and datasets that inform the development of *BPAlign*.

1.2.1 Artificial Neural Networks for Blood Pressure Prediction

Artificial neural networks (ANNs) have been widely applied in the prediction of hypertension and blood pressure patterns due to their ability to model non-linear and complex relationships in clinical data. López-Martínez et al. [1] proposed an ANN-based model trained on the National Health and Nutrition Examination Survey (NHANES) dataset to predict hypertension risk. Their study highlighted the robustness of ANN models in handling heterogeneous health records, capturing hidden interactions among demographic, behavioral, and clinical variables.

The results demonstrated that ANNs could achieve high predictive accuracy in identifying hypertension risk, outperforming conventional statistical methods in many cases. Importantly, the study emphasized the potential of ANN models to be incorporated into telemedicine platforms, where real-time patient data could be analyzed to provide early warnings and support clinical decision-making. This aligns with the objectives of **BPAlign**, where ANN-driven predictions can be translated into actionable insights for practitioners working with rural and underserved populations.

1.2.2 Machine Learning Models Beyond ANN

While artificial neural networks (ANNs) remain a powerful tool for predicting hypertension, recent studies highlight the utility of a broader array of machine learning methods.

Sundström and Schön [2] reflect on the emergence of machine learning-based risk prediction models in cardiovascular medicine. They specifically discuss the widespread use of the Framingham Risk Score as a benchmark for developing predictive models. This underscores the relevance of the Framingham Heart Study dataset, which has been extensively used to train and validate ML models for hypertension and cardiovascular risk. The authors emphasize that machine learning approaches leveraging such well-established datasets can outperform traditional statistical methods by handling complex, multidimensional clinical inputs more effectively.

Guo et al. [3] take this a step further by demonstrating a practical application in hypertension screening. Drawing on NHANES data (2003–2016), they built predictive models using eight machine learning algorithms—such as SVM, Random Forest, and GLMNET—integrating environmental chemical exposures as potential predictors. Among these, the Support Vector Machine (SVM) achieved the best performance, with an AUC of approximately 0.822 and balanced accuracy, precision, and recall around 0.70–0.75 [3].

These findings illustrate two important trends:

1. **Model diversity for better performance**: Beyond ANNs, techniques like SVM and ensemble models can offer competitive or superior predictive capacity.

2. **Novel predictors enhance utility**: Incorporating environmental chemical exposures provides additional layers of risk assessment, which is especially pertinent for population-based screening where such modifiable risk factors could be pivotal.

Together, these studies suggest that **BPAlign** could benefit from integrating diverse machine learning frameworks—especially SVM or ensemble approaches—combined with non-traditional predictors, while also leveraging insights from established datasets such as Framingham, to strengthen predictive accuracy and interpretability in rural telemedicine settings.

1.2.3 Clinical Decision Support Systems in Telemedicine

Mackintosh et al. [4] conducted a systematic review on telemedicine systems with integrated clinical decision support (CDS) in critical care. The review highlighted significant decreases in hospital and ICU mortality and reduced lengths of stay in intervention groups. However, the included studies were methodologically weak—driven by high risk of bias and limited generalizability—underscoring the necessity for robust trials to substantiate observed improvements.
In contrast, an earlier study by [Author, Year (IEEE)] [5] advocated a systemic, modeling-based methodology for CDSS design and evaluation, especially in chronic disease scenarios such as diabetes and renal failure. The authors emphasized the importance of formative and summative assessments, stakeholder analysis, and systemic modeling to ensure both contextual relevance and implementation fidelity.
Together, these works suggest a dual imperative for BPAlign: to pursue rigorous evaluation in real-world telemedicine settings and to ground its design in a systematic, stakeholder-aware framework, ensuring both clinical utility and methodological reliability for hypertension screening in rural populations.

1.3 Gap and Contribution of BPAlign

Identified Gaps: Despite the extensive use of machine learning (ML) and artificial neural networks (ANNs) in hypertension prediction, several gaps remain in existing research:

- Most studies rely on well-established datasets such as NHANES or Framingham, which may not capture the health characteristics of underserved or rural populations [1, 2, 3].
- While ANN- and ML-based models achieve high predictive accuracy, many approaches are *black-box* in nature, providing limited interpretability for clinicians [1, 2].
- Existing telemedicine applications integrating clinical decision support systems (CDSS) often focus on critical care or chronic disease management in hospital settings, rather than preventive screening in rural communities [4, 5].

BPAlign Contributions: BPAlign addresses these gaps by offering a hybrid predictive and analytical platform for hypertension screening in telemedicine contexts, with the following contributions:

1. **Population-specific insights:** By leveraging telemedicine health records from rural populations, BPAlign captures local health trends that are often absent in traditional datasets like NHANES or Framingham.

2. **Interpretable predictions:** BPAlign provides not only hypertension risk predictions but also insights into model behavior, feature importance, and anomaly detection, enabling clinicians to understand and trust the output.

3. **Support for preventive screening:** Unlike CDSS primarily designed for hospital or critical care settings, BPAlign is tailored for early identification of high-risk individuals in remote or underserved areas, facilitating timely intervention.

4. **Flexible ML framework:** BPAlign integrates diverse machine learning models, including ensemble approaches of logistic regression, SVM, random forest and others, allowing comparison and selection of the best-performing model for specific datasets.

5. **Research-ready platform:** In addition to individual risk predictions, BPAlign provides aggregated analytics suitable for public health research and healthcare planning in telemedicine programs.

2 Data, and Models

This study leverages both a benchmark cardiovascular dataset and real-world telemedicine records.The predictive system for hypertension risk. Models were trained on the Framingham Heart Study (FHS) dataset and subsequently applied to the Dhaka University Telemedicine Program (DUTP) data to assess generalizability in rural healthcare contexts.

2.1 Dataset: Framingham Heart Study (FHS)

The Framingham Heart Study (FHS) is a landmark longitudinal cohort study initiated in 1948 in Framingham, Massachusetts, USA, aimed at identifying factors contributing to CVD [6]. The original cohort enrolled 5,209 adults free of overt CVD and has since expanded to include offspring and third-generation cohorts. The dataset is publicly available and has been extensively used for risk prediction and validating cardiovascular biomarkers.Luckily, it contains whether the subject has hypertension or not. I used only nine statically important features[7] from the whole dataset upon the availability in telamrdicine or hospital visits.

2.2 Application Dataset: Dhaka University Telemedicine Program (DUTP)

The Dhaka University Telemedicine Program (DUTP) has accumulated approximately 40,000 patient records from rural Bangladesh. Among them only 15462 records contain useful features that were used also in FSH. Unlike FHS, it does not indicate whether the subject has hypertension or not. So, it was used as

an external data set to test the applicability of trained models in real-world, resource-limited settings.

Key Parameters and Units:

Parameter	**Unit / Format**	**Description**
Age	Years	Participant age at examination
Gender	Categorical (M/F)	Participant sex
Blood Pressure	mmHg (systolic/diastolic)	Measured at clinical visits
Blood Glucose	mg/dL	Fasting glucose measurement
Body Mass Index (BMI)	kg/m^2	Derived from height and weight
Smoking Status	Yes/No	Self-reported smoking habit
Diabetes	Yes/No	Diagnosis or fasting glucose $\geq$ 126 mg/dL
HeartRate	Number	beats per minute

These data form the basis for training machine learning models in BPAlign, providing labeled outcomes and longitudinal features suitable for hypertension risk prediction.

2.3 Data Preprocessing

- Missing values were handled via imputation or removal depending on feature importance.
- Continuous variables (e.g., age, BMI, blood pressure) were standardized.
- Categorical variables (gender, smoking status, medications) were encoded as numeric labels.
- The dataset was split into training (80%) and testing (20%) partitions.
- Unit conversion is needed to use DUTP dataset to keep symmetry with FHS.

2.4 Machine Learning Models

BPAlign integrates multiple supervised learning models for hypertension prediction. This models was chosen with the help of previous analysis on FHS[8, 7, 9] and domain knowledge. Probable best models are:

- **Decision Tree (DT)**
- **Random Forest (RF)**
- **Gradient Boosting Machine (GBM)**
- **Logistic Regression (LR)**
- **Support Vector Machine (SVM)**

Each model produces interpretable outputs including feature importance, model-specific performance reports (JSON), disagreement cases, and ROC curve visualizations.

3 Tools and Methods

3.1 Tools

To implement BPAlign effectively, a combination of modern frontend, backend, and data analysis tools was employed. The chosen stack ensures seamless interaction between user interfaces, model training pipelines, and deployment platforms, while enabling reproducible and interpretable machine learning workflows. The following tools and technologies were utilized:

- **Frontend:** Vite + React JSX + Material-UI (@mui/material)
  Chosen for fast development, modular UI components, and responsive design for interactive dashboards.
- **Backend:** Python (Flask/FastAPI)
  Selected for simplicity, scalability, and easy integration with machine learning pipelines and APIs.
- **Deployment:** PythonAnywhere for backend, GitHub Pages for frontend static assets
  Provides quick and reliable hosting for both dynamic backend services and static frontend assets.
- **Data Analysis:** Kaggle notebooks for preprocessing, visualization, and training pipelines
  Used for exploratory analysis, model prototyping, and reproducible workflows leveraging publicly available datasets.

3.2 System Architechture

The architecture of BPAlign is designed to integrate machine learning pipelines with a lightweight and accessible web interface. It follows a modular client–server approach where the frontend handles user interaction, the backend performs data preprocessing and prediction tasks, and a storage layer manages trained models and analysis artifacts. This separation ensures scalability, reproducibility, and smooth communication between components

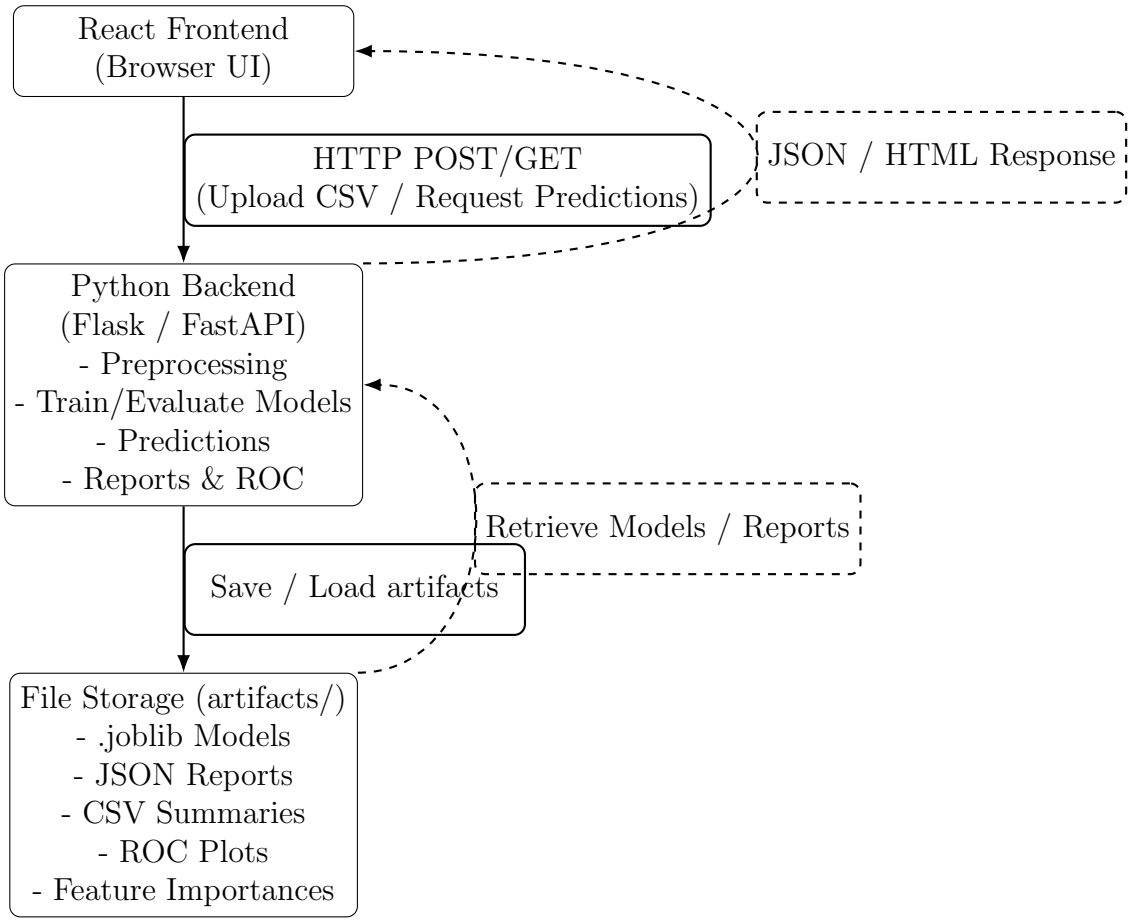


Figure 1: Communication workflow between React frontend, Python backend, and file storage in BPAlign.

3.3 BPAlign Features

Table 1: BPAlign Features and their Functions

Feature	**Function / Description**
Out-of-Sample Prediction	Apply models trained on benchmark datasets (Framingham) to new datasets (e.g., DUTP) to assess predictive performance.
Disagreement Cases	Identify individual cases where different models give conflicting predictions for further review.
Feature Importance	Display key variables influencing predictions for each model.
Model Generalization	Evaluate model performance across datasets to check robustness and transferability.
ROC Curve Visualization	Compare sensitivity and specificity trade-offs across models.
Model Reports	Generate detailed performance metrics (accuracy, recall, precision, F1-score, AUC) and diagnostic comments.
Data Summary Dashboard	Summarize uploaded datasets with descriptive statistics, distributions, and class balance.

3.4 Decision Support Logic

BPAlign includes a lightweight rule-based module that generates human-readable diagnostic comments from training accuracy, test accuracy, and sensitivity. Let gap = train_acc − test_acc. The rules are:

Table 2: Rule-based diagnostic comments used in BPAlign.

Criteria	**Condition**	**Generated Comment**
Generalization gap (gap = train_acc − test_acc)	gap > 0.05	Overfitting (train $\gg$ test).
Generalization gap	gap < -0.01	Possible underfitting (test $>$ train).
Generalization gap	otherwise	Balanced generalization (train $\approx$ test).
Absolute performance (test accuracy)	test_acc ≥ 0.90	High test accuracy.
Absolute performance (test accuracy)	$0.85 \leq$ test_acc < 0.90	Good test accuracy.
Absolute performance (test accuracy)	otherwise	Moderate test accuracy.
Sensitivity (recall for class 1)	$\text{recall}_{\text{pos}} \geq 0.85$	Strong sensitivity for positives (class 1).
Sensitivity (recall for class 1)	$0.75 \leq \text{recall}_{\text{pos}} < 0.85$	Reasonable sensitivity for positives.
Sensitivity (recall for class 1)	otherwise	Low sensitivity for positives—may miss cases.

4 Results

This section presents the outcomes of applying the BPAlign framework to the task of hypertension prediction. The focus is on evaluating model performance on both benchmark and real-world telemedicine data, followed by an exploration of interpretability features and system-level decision support. Through these results, we demonstrate not only the predictive capability of the machine learning models, but also how BPAlign enhances their practical value by offering transparency, generalization checks, and clinically meaningful insights.

4.1 Model Performance

The predictive performance of the machine learning models integrated in BPAlign was evaluated using the Framingham dataset for training and testing, with accuracy and recall serving as the primary metrics. These results provide a baseline understanding of how well each model captures hypertension risk patterns. In addition, performance on the independent DUTP telemedicine

dataset highlights the capacity of the models to generalize to real-world rural health data.

4.1.1 Within-sample performance (benchmark dataset, Framingham)

The following tables summarize the performance of each machine learning model on the test set, highlighting class-wise and overall metrics.

Table 3: Performance of Machine Learning Models on the Framingham Dataset

Model	**Accuracy**	**Precision (Macro)**	**Recall (Macro)**	**F1-score (Ma**
Decision Tree	0.861	0.839	0.834	0.836
Random Forest	0.891	0.870	0.877	0.874
Logistic Regression	0.887	0.863	0.883	0.872
Gradient Boosting	0.892	0.874	0.875	0.874
Support Vector Machine	0.881	0.862	0.861	0.861

Note: Macro-averaged metrics treat both classes equally, ensuring hypertensive patients (minority class) are fairly represented, which is clinically important to avoid missing high-risk individuals.

Table 4: Performance of Machine Learning Models on the Framingham Dataset (Weighted Metrics)

Model	**Accuracy**	**Precision (Weighted)**	**Recall (Weighted)**	**F1-scor**
Decision Tree	0.861	0.860	0.861	
Random Forest	0.891	0.892	0.891	
Logistic Regression	0.887	0.892	0.887	
Gradient Boosting	0.892	0.892	0.892	
Support Vector Machine	0.881	0.881	0.881	

Note: Weighted-averaged metrics account for class imbalance by giving more influence to the majority class. This provides a realistic estimate of overall model performance in practice, ensuring that predictions remain reliable across the full dataset distribution.

Table 5: Per-class precision, recall, and F1-score for all models. Class 0 = non-hypertensive, Class 1 = hypertensive.

Model	Class 0 (Non-Hypertensive)			Class 1 (Hypertensive)			Accuracy	Support
	Precision	Recall	F1	Precision	Recall	F1		
Decision Tree	0.886	0.867	0.876	0.717	0.753	0.735	0.831	848
Random Forest	0.925	0.903	0.913	0.794	0.837	0.815	0.882	848
Logistic Regression	0.896	0.926	0.911	0.823	0.760	0.791	0.875	848
Gradient Boosting	0.934	0.899	0.916	0.793	0.859	0.825	0.887	848
SVM	0.897	0.920	0.908	0.810	0.764	0.787	0.871	848

The table below summarizes the key performance metrics, AUC, and diagnostic comments for all models evaluated on the test set.
The table below summarizes the key performance metrics, AUC, and diagnostic comments for all models evaluated on the test set.

Model	Train Accuracy	Test Accuracy	AUC	Reason
Random Forest	1.000	0.882	0.948	`train >> test`: Overfitting. Goo accuracy. Reasonable sensitivity fo tives.
Decision Tree	1.000	0.831	0.810	`train >> test`: Overfitting. Mo test accuracy. Reasonable sensitivi positives.
Logistic Regression	0.883	0.875	0.948	`train` $\approx$ `test`: Balanced generaliz Good test accuracy. Reasonable se ity for positives.
Gradient Boosting	0.917	0.887	0.952	`train` $\approx$ `test`: Balanced generaliz Good test accuracy. Strong sensitiv positives (class 1).
SVM	0.897	0.871	0.946	`train` $\approx$ `test`: Balanced generaliz Good test accuracy. Reasonable se ity for positives.

Table 6: Summary of training and test performance, AUC, and diagnostic comments for each machine learning model.

4.2 Ensemble Framework

4.2.1 3.1 Model Loading and Weighting

Each candidate model was first loaded from the trained catalog and evaluated based on its test performance. To balance discrimination ability and accuracy, we defined a hybrid score as the average of the area under the ROC curve (AUC) and the test accuracy. Formally, the score for model m is

$$\text{score}_m = \frac{\text{AUC}_m + \text{Acc}_m^{\text{test}}}{2}.$$

The hybrid scores were then normalized to obtain model weights that sum to one:

$$w_m = \frac{\text{score}_m}{\sum_{j=1}^{M} \text{score}_j}.$$

4.2.2 3.2 Ensemble Prediction Strategy

Predictions were generated using a weighted ensemble approach. For an input case x, the probability of belonging to class 1 was computed as the weighted sum of probabilities from all models:

$$P_{\text{ens}}(x) = \sum_{m=1}^{M} w_m \, p^{(m)}(x).$$

The final ensemble prediction was determined by thresholding this probability at 0.5:

$$\hat{y}(x) = \begin{cases} 1 & \text{if } P_{\text{ens}}(x) \geq 0.5 \\ 0 & \text{otherwise.} \end{cases}$$

To quantify confidence, we measured how far the ensemble probability lies from the decision boundary (0.5). This was scaled between 0 and 1 as follows:

$$C(x) = |P_{\text{ens}}(x) - 0.5| \times 2.$$

4.3 Summary on Out-of-sample (cross-dataset).

When a clean and properly formatted dataset, similar to the Framingham dataset, is uploaded, the system generates predictions as summarized in Figure 2.

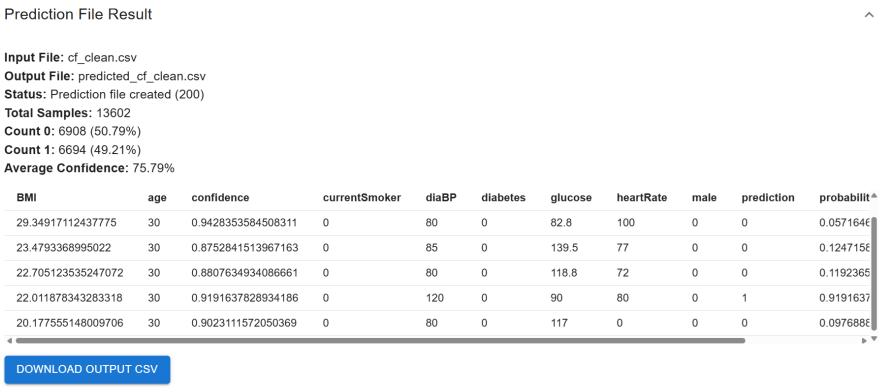


Figure 2: Summary of prediction results on the uploaded dataset.

4.4 Feature Importance

To understand which variables most strongly influence model predictions, feature importance scores were computed for each machine learning model. These scores quantify the relative contribution of each input feature toward predicting hypertension, helping clinicians and researchers interpret model behavior and assess the most critical factors in patient risk assessment.

Feature	Random Forest	Decision Tree	Logistic Regression	Gradient Boosting
Male	0.012	0.006	0.205	0.003
Age	0.076	0.052	0.304	0.017
Current Smoker	0.010	0.010	0.025	0.0003
Diabetes	0.002	0.001	0.044	0.0007
Systolic BP	0.438	0.626	2.445	0.802
Diastolic BP	0.258	0.109	0.989	0.126
BMI	0.090	0.085	0.281	0.027
Heart Rate	0.054	0.042	0.063	0.008
Glucose	0.059	0.070	0.006	0.016

Table 7: Feature importance scores for different machine learning models used in BPAlign. Higher values indicate greater influence on the prediction of hypertension.

To provide interpretability, feature importance was extracted from each individual model. For tree-based models, we relied on the mean decrease in impurity (MDI), which measures how much each feature contributes to reducing Gini or entropy across splits:

$$I(f_j) = \frac{1}{T} \sum_{t=1}^{T} \frac{\Delta\text{Impurity}(f_j, t)}{\text{Total Impurity}(t)}.$$

For linear models, importance was defined as the absolute magnitude of the learned coefficient:

$$I(f_j) = |w_j|.$$

This captures the strength of association between the feature and the prediction, independent of the direction.

4.5 Disagreement Cases

A small subset of cases where the trained models produced conflicting predictions is presented below. Disagreement cases highlight scenarios where different models may interpret the same patient features differently, emphasizing the importance of model comparison and careful clinical interpretation.

4.5.1 3.4 Disagreement Analysis

To assess model diversity, we computed disagreement scores across predictions. For each case i, disagreement was defined as one minus the fraction of models that agreed with the majority class:

$$D_i = 1 - \frac{\max(n_{i,0}, n_{i,1})}{M},$$

Row ID	True Label	Male	Age	Smoker	Diabetes	SysBP	DiaBP	BMI	HR	Glucose
98	1	0	67	0	0	139	80	25.75	75	87
130	1	0	59	1	0	141	86	25.97	70	86
185	0	1	42	0	0	144	78	23.75	62	97
192	0	0	52	0	0	126.5	93	26.79	75	65

Row ID	RF	DT	LR	GB	SVM
98	0	0	0	0	1
130	1	1	0	1	0
185	1	1	0	0	0
192	0	1	0	0	0

Table 8: Selected disagreement cases showing patient features and predictions from Random Forest (RF), Decision Tree (DT), Logistic Regression (LR), Gradient Boosting (GB), and SVM models.

where $n_{i,0}$ and $n_{i,1}$ are the counts of models predicting class 0 and 1, respectively. The average disagreement across all N cases was then calculated as

$$\overline{D} = \frac{1}{N} \sum_{i=1}^{N} D_i.$$

For interpretability, individual disagreement scores were categorized into three levels:

$$\text{Low: } D_i < 0.2, \quad \text{Medium: } 0.2 \leq D_i < 0.5, \quad \text{High: } D_i \geq 0.5.$$

4.6 ROC Curve Comparison

To evaluate and compare the discriminative performance of all trained models, we plotted the Receiver Operating Characteristic (ROC) curves on the test set. The ROC curve illustrates the trade-off between sensitivity (true positive rate) and 1-specificity (false positive rate) across different thresholds. The Area Under the Curve (AUC) provides a single scalar metric summarizing each model's ability to distinguish between hypertensive and non-hypertensive cases.

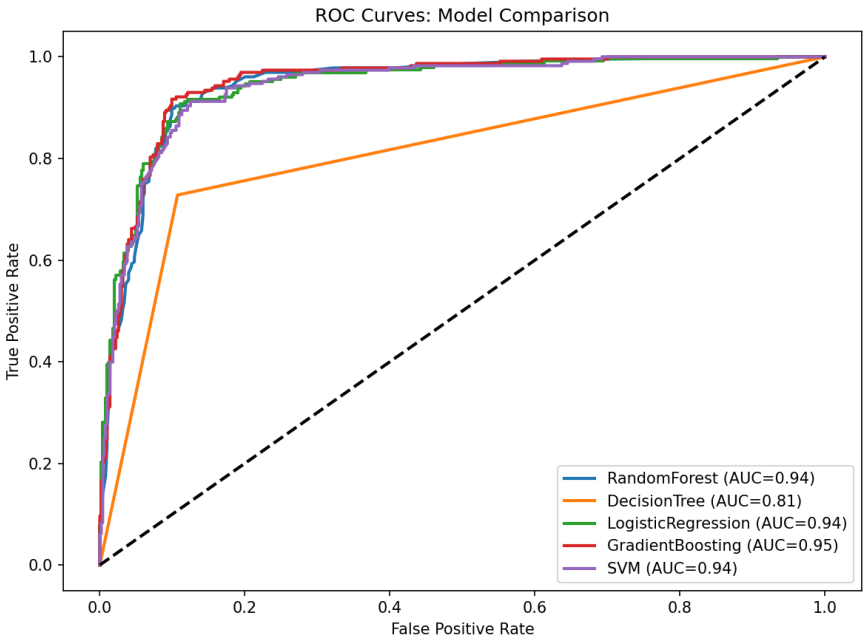


Figure 3: ROC curves for all models evaluated on the test set. The AUC values indicate overall discriminative performance, with higher values corresponding to better separation of classes.

In order to ensure transparency and reproducibility, we formally define the mathematical formulations that govern the proposed ensemble framework. These definitions provide a precise description of how models are loaded and weighted, how predictions and confidence are aggregated, how feature importance is extracted across different model families, and how disagreement among models is quantified. The following subsections describe each component in detail.

Table 9: Mathematical definitions used in the ensemble framework.

Concept	Equation and Explanation
Weighted ensemble probability	$P_{\text{ens}}(x) = \sum_{m=1}^{M} w_m \, p^{(m)}(x)$ where $p^{(m)}(x)$ is the probability for class 1 from model m, and w_m is the normalized model weight.
Ensemble prediction	$\hat{y}(x) = \begin{cases} 1 & \text{if } P_{\text{ens}}(x) \geq 0.5 \\ 0 & \text{otherwise} \end{cases}$ Final class is chosen by thresholding the ensemble probability.
Ensemble confidence	$C(x) = \lvert P_{\text{ens}}(x) - 0.5 \rvert \times 2$ Measures how far the ensemble probability is from indecision (0.5). Higher = more confident.
Model weight calculation	$w_m = \frac{s_m}{\sum_{j=1}^{M} s_j}, \quad s_m = \frac{\text{AUC}_m + \text{Acc}_m^{\text{test}}}{2}$ Weights are proportional to the average of AUC and test accuracy.
Feature importance (tree-based)	$I(f_j) = \frac{1}{T} \sum_{t=1}^{T} \frac{\text{impurity reduction due to } f_j \text{ in tree } t}{\text{total impurity reduction in tree } t}$ Mean decrease in impurity (Gini/entropy) across trees.
Feature importance (linear)	$I(f_j) = \lvert w_j \rvert$ Absolute value of the coefficient w_j, capturing strength of influence regardless of sign.
Disagreement score (per case)	$D_i = 1 - \frac{\max(n_{i,0}, n_{i,1})}{M}$ where $n_{i,0}$ and $n_{i,1}$ are counts of models predicting class 0 and 1, and M is the number of models. $D_i = 0$ if all agree, higher values mean more disagreement.
Average disagreement	$\overline{D} = \frac{1}{N} \sum_{i=1}^{N} D_i$ Mean disagreement score across N test cases.
Disagreement levels	Low: $D_i < 0.2$, Medium: $0.2 \leq D_i < 0.5$, High: $D_i \geq 0.5$ Categorization thresholds for interpretability.
Best model selection	$\text{BestModel}(x) = \arg\max_m \, p^{(m)}(x)$ The single model giving the highest class probability for a given case.

5 System Implementation and Screenshots

The following figures illustrate key interfaces and development views of BPAlign, including some measurements of data quality, feature importance, disagreement cases, model reports, dataset prediction summary, and the deployed prediction interface.

Figure 3 displays the Feature Importance tab of BPAlign, highlighting the key patient attributes that most strongly influence hypertension predictions across models.

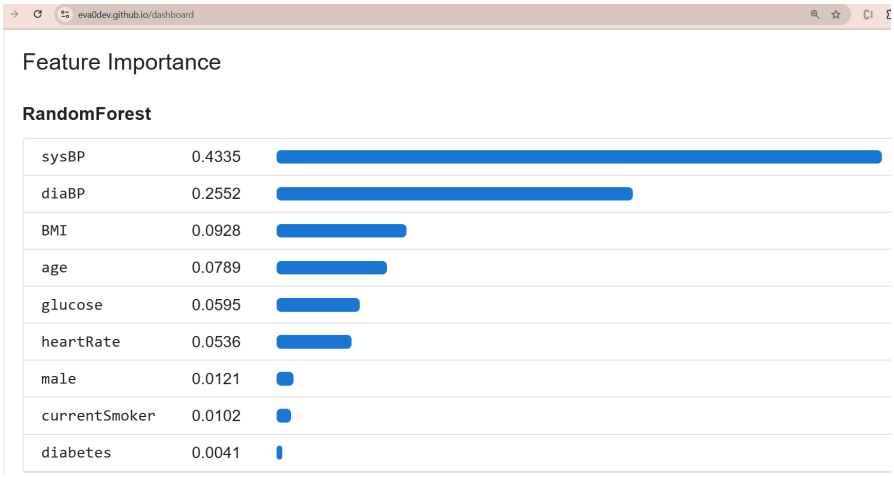


Figure 4: Feature Importance tab in BPAlign showing key variables influencing predictions.

Figure 4 shows a subset of disagreement cases where different models provide conflicting predictions, emphasizing the need for careful model comparison and clinical interpretation.

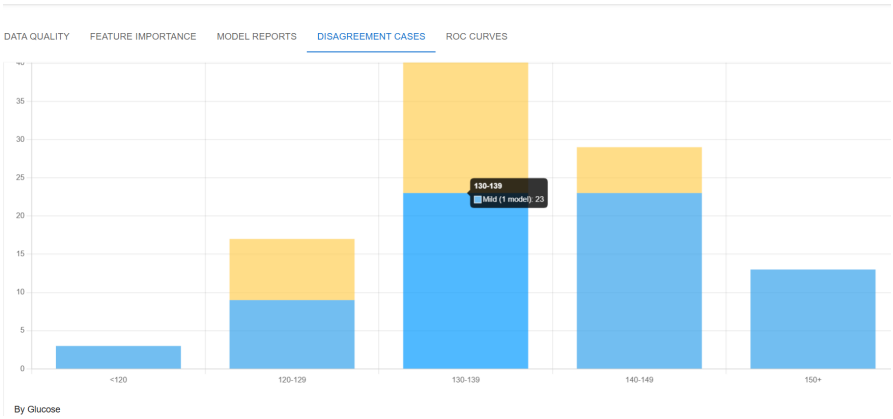


Figure 5: Disagreement cases across models highlighting where predictions differ.

Figure 5 presents a summary of the DUTP dataset, including descriptive statistics, distributions, and class balance, to facilitate a clear understanding of the input data used for prediction.

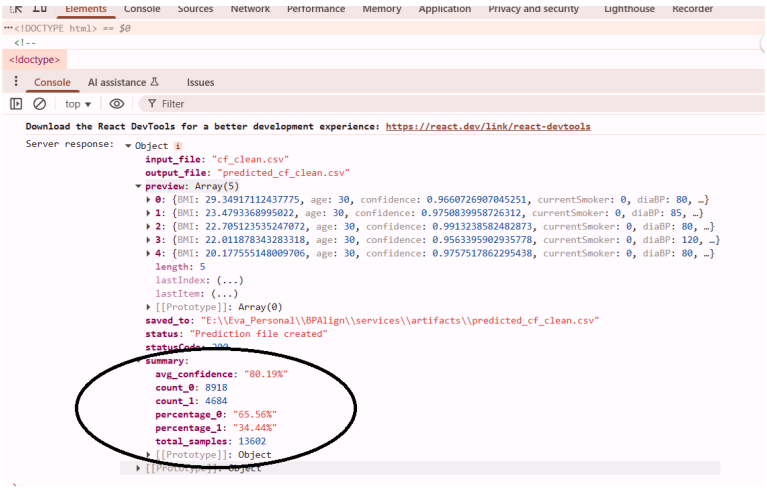


Figure 6: DUTP dataset prediction summary with descriptive statistics and class balance.

Figure 6 illustrates the detailed reports generated for each machine learning model, including metrics such as accuracy, recall, F1-score, and diagnostic comments to support interpretability.prediction.

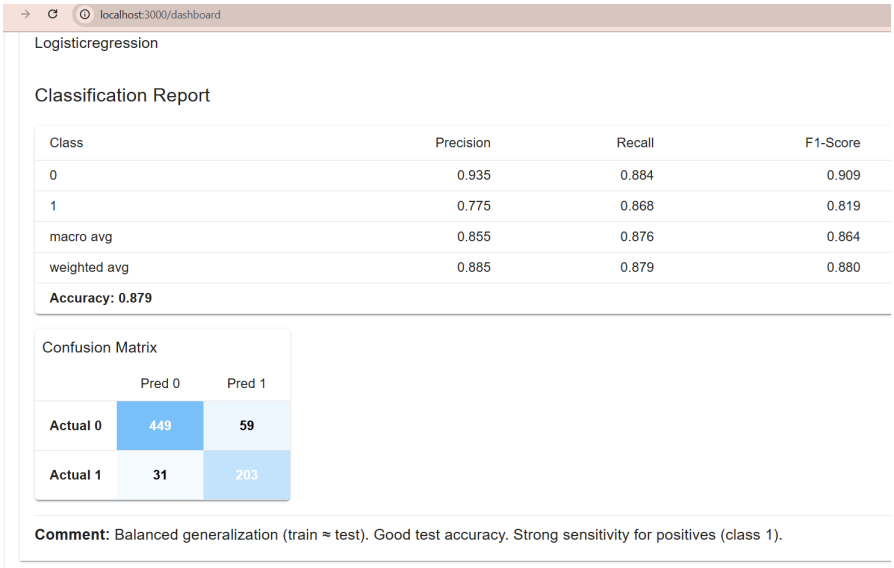


Figure 7: Reports generated for individual models, including accuracy, recall, F1-score, and diagnostic comments.

Figure 7 depicts the deployed BPAlign prediction interface on PythonAnywhere and GitHub Pages, showing how users can input data and obtain hypertension risk predictions in real-time.

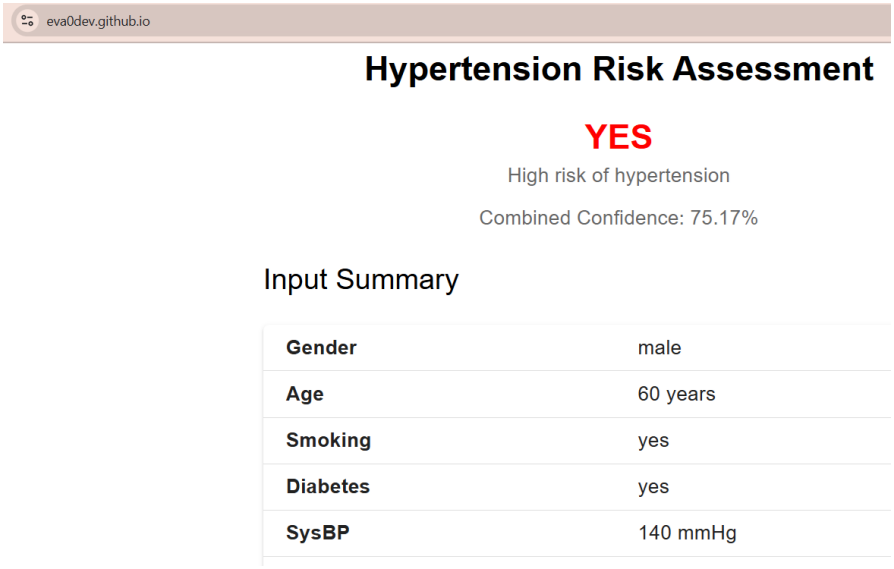


Figure 8: Deployed prediction interface on PythonAnywhere and GitHub Pages.

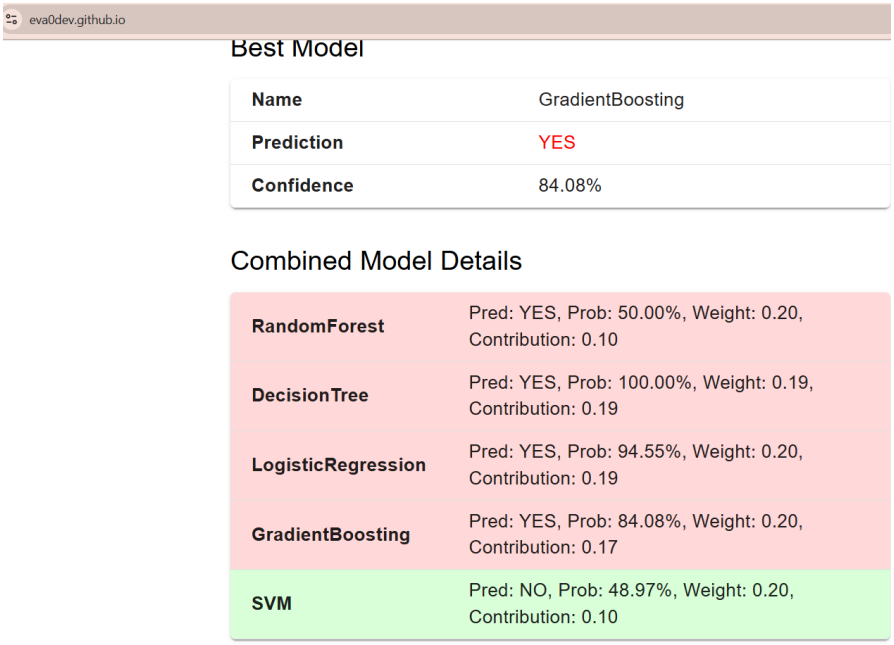


Figure 9: Showing all models resul in Deployement.

6 Discussion

In this section, we discuss the role of different measurements and analyses in ensuring the robustness, reliability, and interpretability of the developed decision support system. By examining data quality, feature importance, model disagreements, and model performance reports, we can better understand how well the system is functioning and whether the models are aligned with the underlying data.

6.1 Data Quality Assessment

Data quality plays a crucial role in determining the effectiveness of any machine learning system. Poor quality data can lead to biased models, unreliable predictions, and weak generalizability. The following aspects are important:

- **Missing values:** Identifying and handling missing values ensures that the model is not trained on incomplete or misleading information.
- **Outliers:** Number of outliers also represents the quality of the data. It also indicates where we should pay extra attention and show the way how to handle these outliers.
- **Duplicates:** Removing duplicate entries avoids overrepresentation of certain samples.
- **Class balance:** Assessing class distribution helps identify imbalance, which may otherwise bias the model towards the majority class.It also help us to make decisions about model types.

In summary, maintaining sound data quality provides a strong foundation for training models, ensuring that the system remains consistent and trustworthy.

6.2 Feature Importance

Understanding feature importance is vital to interpretability and transparency. It allows us to confirm that the model is relying on meaningful attributes rather than noise or spurious correlations. Key points include:

- **Interpretability:** Helps stakeholders understand which variables most influence predictions.
- **Validation:** Confirms whether the most important features align with domain knowledge.
- **Monitoring:** Changes in feature importance over time can signal shifts in data or concept drift.

Thus, feature importance ensures that the model remains explainable, consistent with domain knowledge, and adaptable to changing environments.

6.3 Disagreement Cases

Examining disagreement cases between models highlights where predictions diverge and provides insights into uncertainty. These cases are important for:

- **Error analysis:** Identifying instances where models fail to agree reveals weaknesses in decision boundaries.
- **Reliability:** Frequent disagreements may indicate data ambiguities or the need for further model refinement.
- **Human-in-the-loop:** Disagreement cases can be flagged for expert review to improve overall decision-making.

To conclude, analyzing disagreements allows us to identify weaknesses, reduce risks, and improve the collaborative strength of ensemble models.

6.4 Model Reports

Model performance reports provide a comprehensive view of how well individual models and ensembles perform. They are critical for:

- **Comparative evaluation:** Metrics like accuracy, precision, recall, and F1-score help benchmark multiple models.
- **Weighted contributions:** Understanding how each model contributes to the ensemble improves trust in aggregated results.
- **Consistency check:** Performance reports verify that training and testing results align without overfitting.

In essence, detailed reporting offers assurance that the decision support system is functioning reliably, fairly, and in alignment with real-world needs.
By systematically assessing data quality, feature importance, disagreement cases, and model performance, we ensure that the decision support system is robust and interpretable. These measurements collectively establish confidence that both data and models are well-aligned, enabling reliable predictions and sustainable deployment.

7 Conclusion

BPAlign represents a practical, interpretable, and research-ready tool for hypertension detection in telemedicine and healthcare research. By combining multiple supervised learning models, transparent reporting, and user-friendly interfaces, BPAlign empowers clinicians and researchers to assess hypertension risk, compare model performance, and gain insights from both benchmark and real-world datasets.
The system's dual focus—supporting clinical decision-making and enabling research—positions it as a valuable resource for rural healthcare programs like DUTP. Key contributions include population-specific insights from rural patient data, interpretable predictions through feature importance and diagnostic comments, and flexibility in machine learning model selection and comparison.

7.1 Limitations and Future Work

Despite its strengths, BPAlign has several limitations. First, the DUTP dataset lacks explicit hypertension labels, which constrains the ability to fully validate out-of-sample model performance. Second, some patient features available in benchmark datasets (e.g., Framingham) are missing in the local telemedicine records, potentially limiting predictive accuracy. Third, the system currently relies on static models trained on historical data, which may not fully capture evolving population health trends.
Future work will focus on several areas to address these limitations:

- Continuous model retraining using newly labeled local data to improve generalization.
- Integration of additional patient features and environmental or lifestyle factors to enhance prediction accuracy.
- Expansion of the platform to support other cardiovascular risk predictions beyond hypertension.
- Implementation of automated feedback loops from clinicians to refine diagnostic comments and improve decision support.

By addressing these areas, BPAlign aims to provide even more accurate, context-aware, and actionable insights, ultimately supporting preventive healthcare and better cardiovascular outcomes in underserved populations.

8 Data and Code Availability

The resources for this project are publicly available as follows:

- **Web backend scripts (deployed on PythonAnywhere):** GitHub repository https://github.com/eva0dev/BPAlign-Service.
- **UI code (deployed on GitHub Pages):** Live site https://eva0dev.github.io/BPAlign-UI/ and source code repository https://github.com/eva0dev/BPAlign-UI.
- **Dataset cleaning and preprocessing workflows:** Kaggle notebook https://www.kaggle.com/code/evamajumder/framingham-for-hypertension-detection.

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